Dart XIAO ESP32S3 wiring cheat sheet

Power loop plug out before any wire moves or USB goes in. Pins from `firmware/dart/WIRING.md` (v4, DRV#3 removed; that file wins if this page is older). Full wire list: `firmware/dart/wiring/wire_cut_list.csv`.

XIAO pin	Goes to	Motor
D0 / D1	DRV#1 AIN1 / AIN2	deploy N20
D2 / D3	DRV#1 BIN1 / BIN2	kick N20
D8 / D9	DRV#2 AIN1 / AIN2	both drill motors
D6 / D7	DRV#2 BIN1 / BIN2	foot N20 (these two pins twitch at reset: harmless on the slow worm)
D4 / D5	INA219 + BNO085 SDA / SCL	I2C
3V3	DRV#1 SLP, DRV#2 SLP, INA219 VCC, BNO085 VIN	logic power

Power: 2 x 18350 in series (2S, 6.0 to 8.4 V), OTOTEC BMS, JST-XH loop plug = ON switch, INA219 VIN+ then the motor rail to DRV#1 VM and DRV#2 VM (no buck for motors). MP1584 #1 at 5.04 V taps INA219 VIN+: OUT through a Schottky diode (stripe toward the XIAO) to XIAO 5V, and to the VTX and camera. Capacitor across MP1584 OUT (long leg OUT+).

DRV8833: 3V3 only to the SLP / nSLEEP / EEP pad (check it does NOT beep to VM). DRV#2 channel A drives both drills (right drill leads swapped); channel B drives the foot. No bridge between pads.

Reverse a motor: swap its two input wires at the driver (drills D8 / D9, foot D6 / D7, deploy D0 / D1, kick D2 / D3). One drill wrong only: swap that drill's motor leads. VTX antenna on before any power.